BEING A QUASI-COMPACT IMMERSION IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The pinned source says 'We have to show that $X \to Z$ is a closed immersion' at proof line 1959 where a quasi-compact immersion is meant; delivered verbatim per policy, flagged here for the reviewer.

Allowed standard prerequisites (editorial).

- Framework lemma for descending properties of morphisms. Stacks Project, Descent on Spaces, Lemma lemma-descending-properties-morphisms. Supplies the criterion structure that the proof verifies.
- Closed immersions are \'etale local on the base. Stacks Project, Morphisms of Spaces, Lemma spaces-morphisms-lemma-closed-immersion-local. Supplies part of assumptions (1) and (2).
- Quasi-compact morphisms are \'etale local on the base. Stacks Project, Morphisms of Spaces, Lemma spaces-morphisms-lemma-quasi-compact-local. Supplies the rest of assumptions (1) and (2).
- fpqc descent of quasi-affineness. Stacks Project, Descent on Spaces, Lemma lemma-descending-property-quasi-affine. Makes X a scheme with $X \to Z$ quasi-affine.
- fpqc descent of quasi-compact immersions for schemes. Stacks Project, Descent, Lemma descent-lemma-descending-property-quasi-compact-immersion. Concludes $X \to Z$ is a quasi-compact immersion.

Source reference. Original lemma 041J: sources/stacks-chow/tags.txt, source lines 5294–5294. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03M4: sources/stacks-chow/tags.txt, source lines 4736–4736. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03KG: sources/stacks-chow/tags.txt, source lines 4676–4676. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0423: sources/stacks-chow/tags.txt, source lines 5314–5314. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02L8: sources/stacks-chow/tags.txt, source lines 3399–3399. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:spaces-descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is a quasi-compact immersion" is fpqc local on the base.*

Proof. We will use Lemma 041J to prove this. Assumptions (1) and (2) of that lemma follow from Morphisms of Spaces, Lemmas 03M4 and 03KG. Consider a cartesian diagram

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow & & \downarrow \\ Z' & \longrightarrow & Z \end{array}$$

of algebraic spaces over S where $Z' \to Z$ is a surjective flat morphism of affine schemes, and $X' \to Z'$ is a quasi-compact immersion. We have to show that $X \to Z$ is a closed immersion. The morphism $X' \to Z'$ is quasi-affine. Hence by Lemma 0423 we see that X is a scheme and $X \to Z$ is quasi-affine. It follows from Descent, Lemma 02L8 that $X \to Z$ is a quasi-compact immersion as desired. □